

MATLAB

matlab.mathworks.com

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FILE

MATLAB Drive

```

6 fs = 4000; % Sampling frequency (Hz)
7 N = 50; % Filter order (even number preferred)
8 wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
9 %% --- 1. FIR Low Pass using Rectangular Window ---
10 b_rect = fir1(N, wc, 'low', rectwin(N+1));
11 disp('--- Rectangular Window Coefficients ---');
12 disp(b_rect);
13 %% --- 2. FIR Low Pass using Hamming Window ---
14 b_hamm = fir1(N, wc, 'low', hamming(N+1));
15 disp('--- Hamming Window Coefficients ---');
16 disp(b_hamm);
17 %% --- 3. FIR Low Pass using Hanning Window ---
18 b_hann = fir1(N, wc, 'low', hanning(N+1));
19 disp('--- Hanning Window Coefficients ---');
20 disp(b_hann);
21 %% --- Plot Frequency Responses ---
22 figure;
23 freqz(b_rect,1);
24 title('Low-pass FIR using Rectangular Window');
25 figure;
26 freqz(b_hamm,1);
27 title('Low-pass FIR using Hamming Window');
28 figure;
29 freqz(b_hann,1);
30 title('Low-pass FIR using Hanning Window');

```

Command Window

New to MATLAB? See resources for [Getting Started](#).

```

--- Rectangular Window Coefficients ---
0.0126
0
-0.0137
0
0.0150
0

```

Figure 1: Low-pass FIR using Hanning Window

Figure 2: Low-pass FIR using Hamming Window

Figure 3: Low-pass FIR using Hanning Window

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